Urdu Sarcasm Detection: Models, Experiments, and Error Analysis

Experiment report · binary sarcasm classification on social-media Urdu text · seed 42 · primary metric: macro-F1

1. Data Preparation

The raw dataset (`urdu_sarcastic_dataset.csv`) contained **20,060 rows** with columns `urdu_text` and `is_sarcastic` (plus six unused empty columns). Cleaning proceeded as follows: (i) rows with missing text (105) or missing/invalid labels (56) were dropped, leaving 19,955 rows; (ii) the audit found heavy duplication — 3,646 exact duplicate rows and 4,142 duplicated texts; (iii) most importantly, **597 unique texts carried conflicting gold labels** (the same sentence labeled both sarcastic and non-sarcastic, the same sentence appearing with both label 0 and label 1 in the raw file), so these ambiguous texts were removed entirely and the remainder deduplicated to one copy per text. The final modeling set contains **15,216 unique labeled texts**: 9,037 non-sarcastic (59.39%) and 6,179 sarcastic (40.61%) — a moderately imbalanced but workable distribution.

Exploratory analysis shows sarcastic posts are substantially longer (mean 111.7 chars / median 100) than non-sarcastic ones (mean 71.6 / median 57), and emoji occur frequently in both classes. Text was normalized with a conservative, Urdu-aware pipeline: HTML unescaping, Unicode NFKC, mapping of Arabic-script letter variants (e.g. ة←ه ی←ي ک←ك), removal of tatweel/kashida, and whitespace collapsing. Emoji, punctuation, hashtags, mentions, and Roman-Urdu/code-mixed tokens were deliberately preserved. Data was split **stratified 70/15/15** into train / validation / test (10,650 / 2,283 / 2,283 rows; every split keeps the 59.4% / 40.6% class ratio). The test set was touched exactly once, after model selection on validation.

2. Features and Models

Three model families were trained on identical splits:

- **TF-IDF + Logistic Regression (classical baseline).** Union of word unigram–bigram TF-IDF (min_df=2, max 250k features, sublinear TF) and character 2–5-gram TF-IDF (min_df=2, max 300k features); logistic regression (C=2.0, balanced class weights).
- **Keras BiLSTM (from scratch).** Tokenizer (60k vocab, OOV token), padding to the 95th-percentile train length (49 tokens); Embedding(128) → BiLSTM(64, dropout 0.25) → Dense(64, relu) → Dropout(0.35) → sigmoid; Adam 1e-3, early stopping and LR reduction on validation loss.
- **XLM-RoBERTa (transformer).** Fine-tuning of `xlm-roberta-base` (max 96 tokens), lr 2e-5, batch 2 × grad-accum 8, weight decay 0.01, fp16 + gradient checkpointing, 3 epochs (Tesla T4, 14.56 GiB).

3. Experiments

Ablation 1 — feature type. Character n-grams alone (macro-F1 0.8194) clearly beat word n-grams alone (0.8067); the word+char union was the best classical variant (0.8264). Sub-word patterns capture Urdu morphology and spelling variation that whitespace tokenization misses.

Ablation 2 — emoji. Removing emoji changed validation accuracy only from 0.8327 to 0.8322 (macro-F1 0.8264 → 0.8260): emoji are frequent but carry almost no independent signal — a notable negative result, since social-media sarcasm is often assumed to be emoji-driven.

Training dynamics. The BiLSTM overfits almost immediately (training accuracy 0.9837 by epoch 4 vs. validation ≈0.79, validation loss rising after epoch 1). XLM-RoBERTa improves monotonically (validation macro-F1 0.8129 → 0.8271 → **0.8366**) and was selected by best validation macro-F1; the held-out test set was then evaluated exactly once.

4. Results

Table 1. Validation results (model selection). F1 columns per class; Macro-F1 is the selection metric.

Model (validation)	Acc	P(0)	R(0)	F1(0)	P(1)	R(1)	F1(1)	Macro-F1
TF-IDF word, emoji kept	0.8125	0.8510	0.8296	0.8402	0.7596	0.7875	0.7733	0.8067
TF-IDF char, emoji kept	0.8257	0.8538	0.8525	0.8531	0.7847	0.7864	0.7856	0.8194
TF-IDF word+char, emoji removed	0.8322	0.8580	0.8599	0.8589	0.7944	0.7918	0.7931	0.8260
TF-IDF word+char, emoji kept	0.8327	0.8581	0.8606	0.8594	0.7952	0.7918	0.7935	0.8264
BiLSTM	0.8025	0.8606	0.7965	0.8273	0.7315	0.8112	0.7696	0.7984
XLM-RoBERTa	**0.8406**	**0.8869**	0.8385	**0.8620**	0.7812	**0.8436**	**0.8112**	**0.8366**

Table 2. Held-out test results for all trained models (n = 2,283).

Model (test)	Acc	P(0)	R(0)	F1(0)	P(1)	R(1)	F1(1)	Macro-F1

XLM-RoBERTa (selected)	**0.8200**	**0.8729**	0.8156	0.8433	0.7539	**0.8263**	**0.7885**	**0.8159**
TF-IDF word+char / LogisticRegression	0.8195	0.8565	**0.8363**	**0.8463**	**0.7685**	0.7950	0.7815	0.8139
BiLSTM	0.8108	0.8762	0.7935	0.8328	0.7346	0.8360	0.7821	0.8074

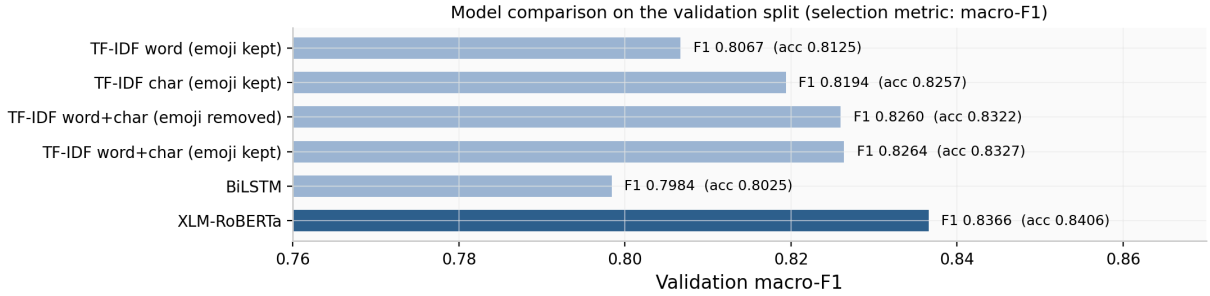


Figure 1. Validation macro-F1 (with accuracy) across all six trained configurations. XLM-RoBERTa wins model selection; the TF-IDF word+char baseline is within 0.011 of it.

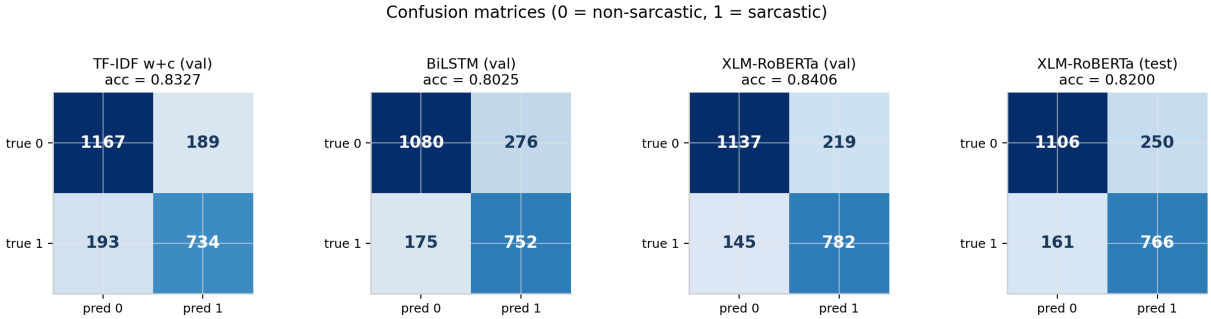


Figure 2. Confusion matrices for the three model families on validation, and for the selected XLM-RoBERTa on the held-out test set (counts derived from the reported per-class precision/recall; class supports 1,356 / 927).

5. Error Analysis

XLM-RoBERTa misclassified **411 of 2,283** test posts (18.0%): 250 false positives (non-sarcasm predicted sarcastic) versus 161 false negatives (missed sarcasm). The asymmetry matches the metrics — sarcasm precision (0.7539) trails sarcasm recall (0.8263). Manual inspection of the misclassified test rows (a 10-example review sheet was exported for annotation) reveals recurring patterns, illustrated below.

Table 3. Representative misclassified test examples (selected model: XLM-RoBERTa). Ellipses mark truncation in the source data; emoji shown in brackets.

#	Test text (Urdu)	Gold → Pred	Error type	Commentary
1	بندو گانے کو ماتا مگر بیل کو باپ نہیں مانتے ہی…	0 → 1	False positive	A jokey proverb-like jab about “Bandu”. The humor register resembles sarcasm, so the model follows surface humor cues; gold says non-sarcastic — a genuine joke-vs-sarcasm boundary case, arguably label noise.
2	او پانڈری تجھے بس نقل نقل اور نقل کرنی ہے آئی…	0 → 1	False positive	A direct taunt (“all you can do is copy, copy, copy”). The ridicule is overt rather than ironic, but mockery shares lexical signals with sarcasm, so the model over-fires on insults.
3	جس دن جنرل صاحب کو ہدایت آے گی اس دن بہت دیر ہ…	1 → 0	False negative	“The day the General comes to his senses, it will already be too late” — implicit irony aimed at a public figure, with no emoji or marker and a truncated ending. Such context-dependent jabs are the classic missed-sarcasm pattern.
4	اس قرار داد میں یوکرین پر روسی حملے کا ذکر نہیں…	0 → 1	False positive	A factual complaint about a resolution’s wording on the Ukraine war. Loaded political topics are over-predicted as sarcastic — a “political criticism ⇒ sarcasm” shortcut.

5	کیڑوں کا بل ہو گا نسل در نسل چلتا آ رہا ہے [laughing ×2]	0 → 1	False positive	A playful quip with laughing emoji, labeled non-sarcastic. Emoji-heavy humor pulls the model toward sarcasm; with the emoji-removal ablation showing no metric change, emoji act as a confound interacting with content rather than a clean signal.
6	اپ آپ سے مذاق نہیں کرنا تو کیا پھپھو سے [laughing ×2, ROFL, see-no-evil ×2]…	0 → 1	False positive	Teasing repartee (“if we can’t joke with you, should we joke with your aunt?”) ending in five emoji. Teasing and sarcasm overlap heavily; the model cannot separate friendly banter from irony.
7	لوگ ہمارے بارے میں کیا سوچتے ہیں یہ بھی ہم سوچ…	1 → 0	False negative	Deadpan self-deprecation (“what people think of us — we think that too”). No lexical exaggeration or emoji; it requires pragmatic inference, which the model misses.
8	ہمارا قومی چور واپس کرو۔	1 → 0	False negative	“Return our national thief” — a sarcastic slogan whose irony depends on world knowledge. A short, context-free post gives the model almost nothing to work with.

Summary of error patterns. (i) *Joke/teasing confound:* humorous but non-sarcastic posts account for many false positives — humor and sarcasm share surface features, and gold labels at this boundary are themselves inconsistent (597 conflicting texts were removed in cleaning). (ii) *Implicit sarcasm:* false negatives are typically deadpan, context-dependent irony with no explicit markers. (iii) *Topic bias:* political criticism is over-predicted as sarcastic. (iv) *Label-noise ceiling:* noisy annotation at the humor–sarcasm boundary likely caps accuracy near the observed ~82%.

6. Conclusion

Fine-tuned XLM-RoBERTa is the best model (validation macro-F1 0.8366; test accuracy 0.8200, macro-F1 0.8159), but the TF-IDF word+char baseline trails it by only 0.002 macro-F1 on test — a remarkably strong, cheap baseline. The from-scratch BiLSTM is weakest (0.8108), crippled by overfitting on 10.6k training texts. Notable negative results: emoji carry almost no independent signal, and character n-grams add substantially over word n-grams. Future directions: larger, cleaner labeled data; context/thread-aware modeling; decision-threshold tuning to convert excess false positives into precision; and re-annotation of humor–sarcasm boundary cases.

Reproducibility: seed 42; stratified 70/15/15 split; artifacts saved from the notebook (tfidf_urdu_sarcasm.joblib, urdu_sarcasm_bilstm.keras, urdu_sarcasm_xlmr, run_metadata.json); primary metric macro-F1. Environment: Google Colab, Tesla T4, Python 3.13.